Variable-coefficient positivity generators on $\mathbb{R}^n$

Literature-already-answered note for arXiv:2308.10455

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Status. Open Problem 6.1 is completely answered by di Dio–Schmüdgen, arXiv:2407.15654, Theorem 5.12. The later theorem treats exactly the same degree-preserving canonical differential operators and gives a necessary-and-sufficient pointwise Lévy–Khintchine description. This packet records that later-literature resolution; it does not claim a new theorem.

The source problem

Let $\mathrm{Pos}(\mathbb{R}^n)$ denote the cone of real polynomials nonnegative on $\mathbb{R}^n$. The source observes that a linear operator on $\mathbb{R}[x_1, \ldots, x_n]$ is degree preserving precisely when it has the canonical representation

$$A = \sum_{\alpha \in \mathbb{N}_0^n} q_\alpha(x)\, \partial^\alpha, \qquad \deg q_\alpha \leq |\alpha|.$$

This condition makes e^{tA} well defined on every finite-degree polynomial subspace. Open Problem 6.1 on source PDF page 22 asks for a description of all such A, without the constant-coefficient restriction, for which

$$e^{tA}\, \mathrm{Pos}(\mathbb{R}^n) \subseteq \mathrm{Pos}(\mathbb{R}^n) \qquad (t \geq 0).$$

group theory. Besides the Open Problem 6.1 below this is another direction where further studies have to be done.

6. Summary and Open Question

If

$$A = \sum_{\alpha\in\mathbb{N}_0^n} q_\alpha \cdot \partial^\alpha \quad \text{with} \quad q_\alpha \in \mathbb{R}[x_1,\dots,x_n]_{\leq|\alpha|} \tag{20}$$

then

$$A\mathbb{R}[x_1,\dots,x_n]_{\leq d} \subseteq \mathbb{R}[x_1,\dots,x_n]_{\leq d} \quad \text{for all} \quad d \in \mathbb{N}_0, \tag{21}$$

i.e., A is called *degree preserving*. In fact for any linear $A : \mathbb{R}[x_1,\dots,x_n] \to \mathbb{R}[x_1,\dots,x_n]$ we have (20) $\Leftrightarrow$ (21). (20) $\Rightarrow$ (21) is clear. For (21) $\Rightarrow$ (20) take in (20) the smallest α with respect to the lex-order such that $\deg q_\alpha > |\alpha|$. Then $\deg Ax^\alpha > |\alpha|$ which is a contradiction to (21).

Similar to Corollary 4.6 we have that the partial differential equation

$$\partial_t p(x,t) = Ap(x,t) \quad \text{with} \quad p(x,0) = p_0(x) \in \mathbb{R}[x_1,\dots,x_n]$$

and A as in (20) is equivalent to a linear system of ordinary differential equations in the coefficients of p. Hence, it has a unique solution $p(\,\cdot\,,t) = \exp(tA)p_0$ for all $t \in \mathbb{R}$, i.e., $\exp(tA)$ is well-defined for any A as in (20) and $t \in \mathbb{R}$. If additionally A is a positivity preserver then $\exp(tA)$ is a degree preserving positivity preserver with polynomial coefficients for all $t \geq 0$. In Main Theorem 4.11 we have already seen for the positivity preservers with constant coefficients that additional A appear. The restriction that A is degree preserving is a natural restriction since otherwise $\exp(tA)\mathbb{R}[x_1,\dots,x_n] \not\subseteq \mathbb{R}[x_1,\dots,x_n]$ for any $t > 0$. So we arrive at the following open problem.

Open Problem 6.1. *What is the description of the set*

$$\left\{ A = \sum_{\alpha\in\mathbb{N}_0^n} q_\alpha \cdot \partial^\alpha \;\middle|\; \exp(tA) \textit{ is a positivity preserver for all } t \geq 0 \right\}$$

of all positivity generators, i.e., not necessarily with constant coefficients?

The constant coefficient case is completely solved by Main Theorem 4.11. Part of the non-constant coefficient cases are covered by the previous case that A is a degree preserving positivity preserver. For the non-constant coefficient cases note that we are then in the framework of a non-commutative infinite dimensional Lie group and its Lie algebra. Here the theory is much richer, see e.g. [Omo74, Kac85, Omo97, SHNW02, Wur04, Sch23] and references therein.

Source evidence: arXiv:2308.10455, PDF page 22 (unaltered crop).

The exact later theorem

Theorem 5.12 of arXiv:2407.15654 (PDF page 25) writes the same operator as

$$A = \sum_{\alpha \in \mathbb{N}_0^n} \frac{a_\alpha(x)}{\alpha!} \partial^\alpha, \qquad a_\alpha \in \mathbb{R}[x_1, \ldots, x_n]_{\le |\alpha|}.$$

It proves that e^{tA} preserves $\operatorname{Pos}(\mathbb{R}^n)$ for every $t \ge 0$ if and only if, for every $y \in \mathbb{R}^n$, there are a positive-semidefinite symmetric matrix $\Sigma(y) = (\sigma_{ij}(y))$, a drift vector $b(y) = (b_1(y), \ldots, b_n(y))^T$, and a measure ν_y on $\mathbb{R}^n$ such that

$$\int_{\|z\|_2 \ge 1} |z_i| \, d\nu_y(z) < \infty, \qquad \int_{\mathbb{R}^n} |z^\alpha| \, d\nu_y(z) < \infty \quad (|\alpha| \ge 2),$$

and the coefficient values obey

$$\begin{aligned} a_{e_i}(y) &= b_i(y) + \int_{\|z\|_2 \ge 1} z_i \, d\nu_y(z), \\ a_{e_i+e_j}(y) &= \sigma_{ij}(y) + \int_{\mathbb{R}^n} z^{e_i+e_j} \, d\nu_y(z), \\ a_\alpha(y) &= \int_{\mathbb{R}^n} z^\alpha \, d\nu_y(z) \qquad (|\alpha| \ge 3). \end{aligned}$$

Here a_0 is an arbitrary real constant, as forced already by $\deg a_0 \le 0$. Thus the zeroth-order scaling, drift, Gaussian part, and jump moments give the requested complete description.

Why this is the same question

Set $a_\alpha = \alpha! q_\alpha$. This is a bijective change of normalization and preserves the bound $\deg a_\alpha \le |\alpha|$. Both papers use the same cone of globally nonnegative polynomials, the same semigroup condition for every $t \ge 0$, and arbitrary polynomial (not merely constant) coefficients.

Feature	Open Problem 6.1	Theorem 5.12
Operator class	$\sum q_\alpha \partial^\alpha$, $\deg q_\alpha \le \lvert\alpha\rvert$	$\sum (a_\alpha/\alpha!)\partial^\alpha$, same bound
Positivity domain	all of $\mathbb{R}^n$	$K = \mathbb{R}^n$
Requested property	e^{tA} positive for all $t \ge 0$	condition (i), verbatim
Requested output	description of every generator	iff Lévy–Khintchine data at every y

Proof intuition

For each point y, freeze the polynomial coefficients:

$$A_y = \sum_{\alpha} \frac{a_\alpha(y)}{\alpha!} \partial^\alpha.$$

The key result behind Theorem 5.12 is the equivalence

$$A \text{ generates a positivity-preserving semigroup} \quad \Longleftrightarrow \quad A_y \text{ does so for every } y \in \mathbb{R}^n.$$

The forward direction uses translation invariance of each constant- coefficient A_y, together with the first-order expansion of e^{tA} and a product-limit argument. The reverse direction constructs positive approximants from the frozen semigroups and again takes a closed product limit. The already-solved constant-coefficient generator theorem then applies pointwise. Its Lévy–Khintchine formula is exactly the matrix, drift, and measure condition displayed above. This explains both why the answer is pointwise in y and why no compatibility conditions beyond the given polynomial coefficient functions are missing.

and since all $G_{t/k}$ are K-positivity preservers also $G_{t/k}^k$ are K-positivity preservers as well. Therefore, since the set of K-positivity preservers is closed by Lemma 4.6 (ii) we have that e^{tA} is a K-positivity preserver. Since $t \geq 0$ was arbitrary we have that (ii) is proved. □

The implication "(ii) ⇒ (i)" in Theorem 5.11 does not hold in general, see e.g. Remark 5.13. For $K = \mathbb{R}^n$ the implication holds and we get the following.

Theorem 5.12. *Let*

$$A = \sum_{\alpha \in \mathbb{N}_0^n} \frac{a_\alpha}{\alpha!} \cdot \partial^\alpha$$

be with $a_\alpha \in \mathbb{R}[x_1, \dots, x_n]_{\leq |\alpha|}$ for all $\alpha \in \mathbb{N}_0^n$. Then the following are equivalent:

(i) $A \in \mathfrak{D}_+$, i.e., e^{tA} is a positivity preserver for all $t \geq 0$.

(ii) For every $y \in \mathbb{R}^n$ there exist a symmetric matrix $\Sigma(y) = (\sigma_{i,j}(y))_{i,j=1}^n$ with real entries such that $\Sigma(y) \succeq 0$, a vector $b(y) = (b_1(y), \dots, b_n(y))^T \in \mathbb{R}^n$, $a_0 \in \mathbb{R}$, and a measure ν_y on $\mathbb{R}^n$ with

$$\int_{\|x\|_2 \geq 1} |x_i| \, \mathrm{d}\nu_y(x) < \infty \qquad \textit{and} \qquad \int_{\mathbb{R}^n} |x^\alpha| \, \mathrm{d}\nu_y(x) < \infty$$

for $i = 1, \dots, n$ and $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \geq 2$ such that

$$a_{e_i}(y) = b_i(y) + \int_{\|x\|_2 \geq 1} x_i \, \mathrm{d}\nu_y(x) \qquad \textit{for } i = 1, \dots, n,$$

$$a_{e_i + e_j}(y) = \sigma_{i,j}(y) + \int_{\mathbb{R}^n} x^{e_i + e_j} \, \mathrm{d}\nu_y(x) \qquad \textit{for all } i, j = 1, \dots, n,$$

and

$$a_\alpha(y) = \int_{\mathbb{R}^n} x^\alpha \, \mathrm{d}\nu_y(x) \qquad \textit{for } \alpha \in \mathbb{N}_0^n \textit{ with } |\alpha| \geq 3.$$

Proof. By Proposition 2.12 we have that (i) is equivalent to the fact that

$$A_y = \sum_{\alpha \in \mathbb{N}_0^n} \frac{a_\alpha(y)}{\alpha!} \cdot \partial^\alpha$$

is a generator of a positivity preserving semi-group with constant coefficients for all $y \in \mathbb{R}^n$.

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Supporting evidence: arXiv:2407.15654, Theorem 5.12 on PDF page 25 (unaltered crop).

Human review. Compare Open Problem 6.1 on source PDF page 22 with Theorem 5.12 on supporting PDF page 25. In particular, verify the harmless normalization $a_\alpha = \alpha! q_\alpha$ and the shared degree bound.

References

[1] P. J. di Dio, *On Positivity Preservers with constant Coefficients and their Generators*, arXiv:2308.10455. See Open Problem 6.1, PDF page 22.

[2] P. J. di Dio and K. Schmüdgen, *K-Positivity Preservers and their Generators*, arXiv:2407.15654. See Theorem 5.12, PDF page 25.